

Due: August 11, 2026

1. Write the following in set notation:
 - a) The set of all real numbers less than 5.
 - b) The set of all real numbers greater than 8, but less than 73.
2. Given the sets: $S_1 = \{2, 4, 6\}$, $S_2 = \{7, 2, 6\}$, $S_3 = \{4, 2, 6\}$ and $S_4 = \{2, 4\}$, find
 - a) $S_1 \cup S_3$
 - b) $S_2 \cap S_4$
 - c) $S_3 \cup S_1 \cup S_4$
 - d) $S_1 \cap S_3 \cap S_4$
3. Enumerate all of the subsets of the set $\{1, 7, 13\}$.
4. Given $S_1 = \{3, 6, 9\}$, $S_2 = \{a, b\}$ and $S_3 = \{m, n\}$, find the Cartesian products:
 - a) $S_2 \times S_3$
 - b) $S_3 \times S_1$
 - c) $S_1 \times S_2 \times S_3$
5. If the *domain* for the function $y = 10 - 4x$ is the set $\{x \mid -2 \leq x \leq 3\}$ find the *range* of the function and express it as a set.
6. If the *domain* for the function $y = 3x$ is the set $\{x \mid x \geq 3\}$ find the *range* of the function and express it as a set.
7. For the function $y = -x^2$, if the domain is the set of all nonnegative real numbers, what will its range be?
8. Graph the following functions:
 - a) $y = 8 + 3x$
 - b) $y = 8 - 3x$
 - c) $y = 3x + 12$
 - d) $y = 4$
 - e) $y = x$
 - f) $y = x^2$
9. Identify the vertical intercept and the slope for the following functions:
 - a) $y = 14 + 6x$

b) $y = -2 - 6x$

c) $y = x$

d) $y = 2$

10. Simplify the following expressions:

a) $x^2 \times x^8$

b) $x^2 \times x^3 \times x^4$

c) $x^3 \times y^3 \times z^3$

d) $(x^4)^5$

e) $(x^a)^3$

f) x^{-2}

g) $\frac{x^3}{x^{-3}}$

h) $\frac{x^{1/2} \times x^{1/3}}{x^{2/3}}$

i) $\frac{x^4}{x^3}$

j) $\frac{(x^{1/3})^3}{x^{1/3}}$